BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, MARDAN

INTERMEDIATE (12TH CLASS) EXAMINATION, 2021

CHEMISTRY

Past Paper 2021 • Paper Code: MRD-XII-21 (A)

Retyped faithfully from the original board paper — same questions, no changes.

Section-A

Time Allowed: 20 Minutes | **Maximum Marks:** 18 | **Paper Code:** MRD-XII-21 (A) VERSION: B

Note: Note: There are three sections in this paper i.e. A, B & C. Instructions: Please attempt Section-A on the MCQs Answer Sheet only. Only black ball point or marker may be used for shading the relevant circle. Please make sure that there is no cutting, over writing, erasing, or multiple circles shading.

Q1. Choose the correct option i.e. A,B,C, or D.

Q1. Addition of an acid catalyst in nucleophilic addition reaction of aldehydes causes .....

(A) Increase in electrophilicity of carbonylic "C"
(B) Increase in nucleophilicity of nucleophile
(C) Reversibility of reaction
(D) Deceleration

Q2. Cholesterol belongs to family of .....

(A) Carbohydrate
(B) Protein
(C) Lipids
(D) Enzymes

Q3. The least reactive alkyl halide is .....

(A) R−F
(B) R−Cl
(C) R−I
(D) R−Br

Q4. Which one of the following is a condensation polymer?

(A) PVC
(B) Polythene
(C) Nylon
(D) None of these

Q5. Which of the following compounds precipitate Cu_2O from fehling reagent but does not react with sodium metal?

(A) C_2H_5OH (B) C_2H_5CHO

(C) CH_3COCH_3 (D) CH_3OCH_3

Q6. Carbohydrates are stored in human body in the form of.....

(A) Glycogen (B) Glucose

(C) Starch (D) Galactose

Q7. Ozone layer in upper atmosphere is being destroyed by .....

(A) Chlorofluorocarbons (B) SO_2

(C) Smog (D) Petrochemical oxidant

Q8. Which one of the following is not a pesticide?

(A) Lead arsenate (B) Parathion

(C) DDT (D) None of these

Q9. Specificity of enzymes depend upon .....

(A) Active site (B) Linear sequence

(C) Turn over number (D) None of these

Q10. Triglycerides, which are liquid at room temperature, are .....

(A) Fats (B) Oils

(C) Waxes (D) All of these

Q11. The temperature of stratosphere is .....

(A) 15°C to −52°C (B) −56°C to −2°C

(C) −52°C to −2°C (D) −2°C to 92°C

Q12. Bronze is the alloy of copper and .....

(A) Sn (B) Zn

(C) Fe (D) Au

Q13. The chemical bond in $BeCl_2$ is ..... bond.

(A) Ionic (B) Covalent

(C) Dative (D) Coordinate

Q14. ..... is an alicyclic organic compound.

(A) Benzene (B) Pyridine

(C) Toluene (D) Cyclobutane

Q15. The numbers of hydrogen in pyridine are .....

(A) 5 (B) 6

(C) 4 (D) None of these

Q16. Alcohols show no chemical reaction with.....

(A) Water (B) Aqueous solution of NaOH

(C) Sodium metal (D) Both A & B

Q17. In primary alkyl halide, the carbon atoms to which the halo group is attached, is in turn bonded directly to..... H-atoms at least.

(A) One (B) Two

(C) Three (D) Four

Q18. Which one of the following deactivates benzene ring and is ortho/para directing?

(A) $-NH_2$ (B) $-F$

(C) $-OH$ (D) None of these

Section-B

Time Allowed: 2:40 Hours | **Maximum Marks:** 40

Q2. Write short answers to any ten (10) of the following parts. All parts carry equal marks.

(i) Why phenol is more acidic than alcohol?

(ii) Describe any two methods of preparation of alkynes.

(iii) Draw the open and cyclic structure of the following compounds: (a) Glucose (b) Fructose (c) Mannose (d) Galactose

(iv) Why NaCl does not conduct electricity in solid state?

(v) What is Lucas test?

(vi) Differentiate between paramagnetic and diamagnetic substances.

(vii) Explain the flame test with example of colour of rubidium (Rb).

(viii) Write down the resonance structure of benzene.

(ix) Why is acid rain considered as a threat to aquatic life and historical monuments?

(x) What are the uses of PVC and Nylon?

(xi) What is meant by primary, secondary and tertiary alcohols?

(xii) Write down chemical properties of Lipids showing their chemical reaction.

(xiii) Write note on dissolved solids.

Section-C

Maximum Marks: 27

Note: Answer any three (3) questions. All questions carry equal marks.

Q3.

(a) How beryllium differs from other members of its group?

(b) What are complex ions? Explain their shape and colours.

Q4.

(a) Write the IUPAC names of the following compounds. (i) HCOOH (ii) CH_3COOH (iii) $CH_3CH_2CH_2COOH$ (iv) CH_3CH_2COOH (v) CH_3CF_2COOH

(b) Explain the acidity of carboxylic acids.

Q5.

(a) Describe the steps while naming Ketones in IUPAC system.

(b) Define inert pair effect. Why Sn and Pb use it?

Q6.

(a) Explain the stabilities of halides of 4th A elements with examples.

(b) Explain at least four characteristics of organic compounds.